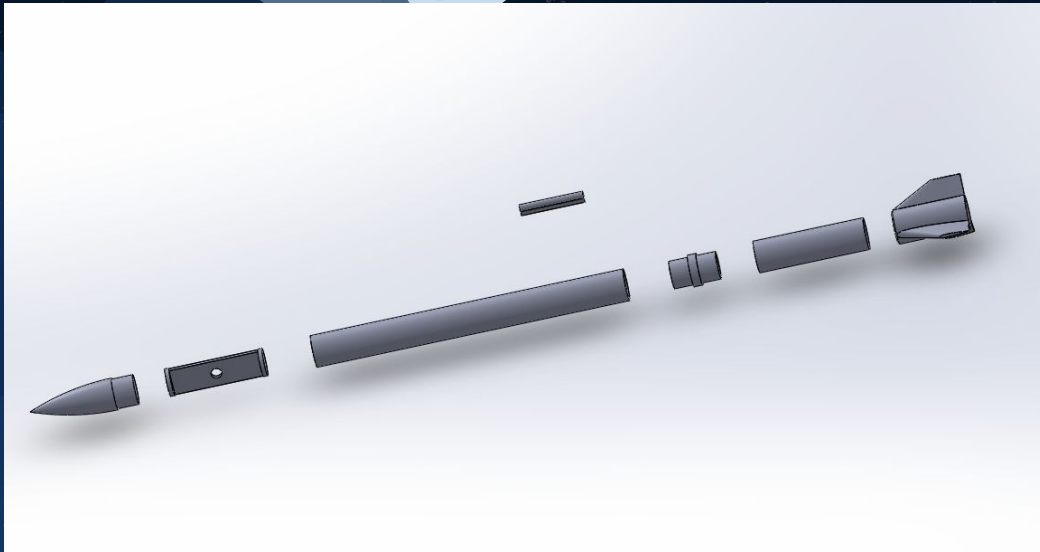


CDR Proxima RPL

Zein Rady, Daniel Park, Cia Kingston,
Achintya Rai, Kaleb Park, Esha Rami



TEAM GOALS



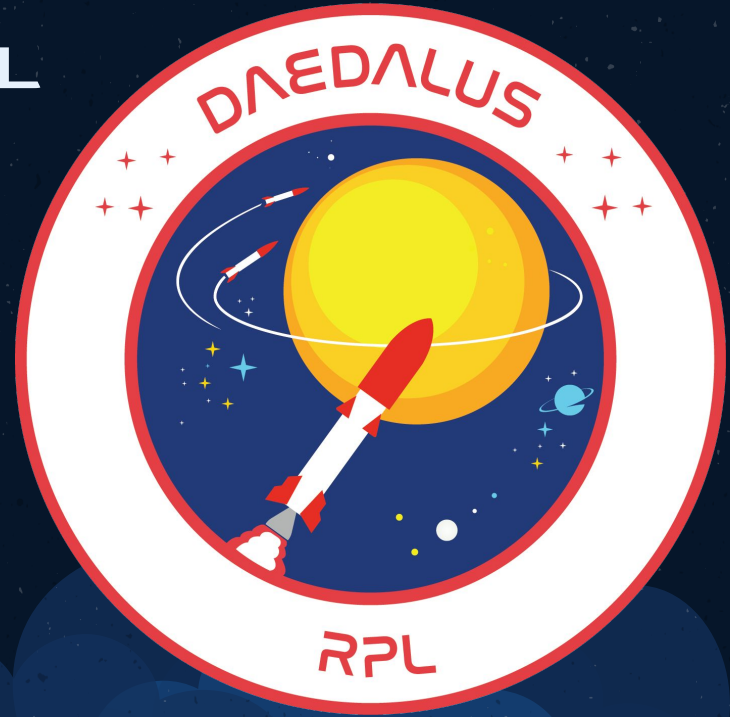
TEAM GOALS

Our team will strive to achieve the following goals:

1. Get hands on experience with avionics (hardware + software)
2. Begin manufacturing and 3D printing rocket
3. Launch and reach an apogee of 3500 ft



OVERVIEW OF PHYSICAL PROPERTIES

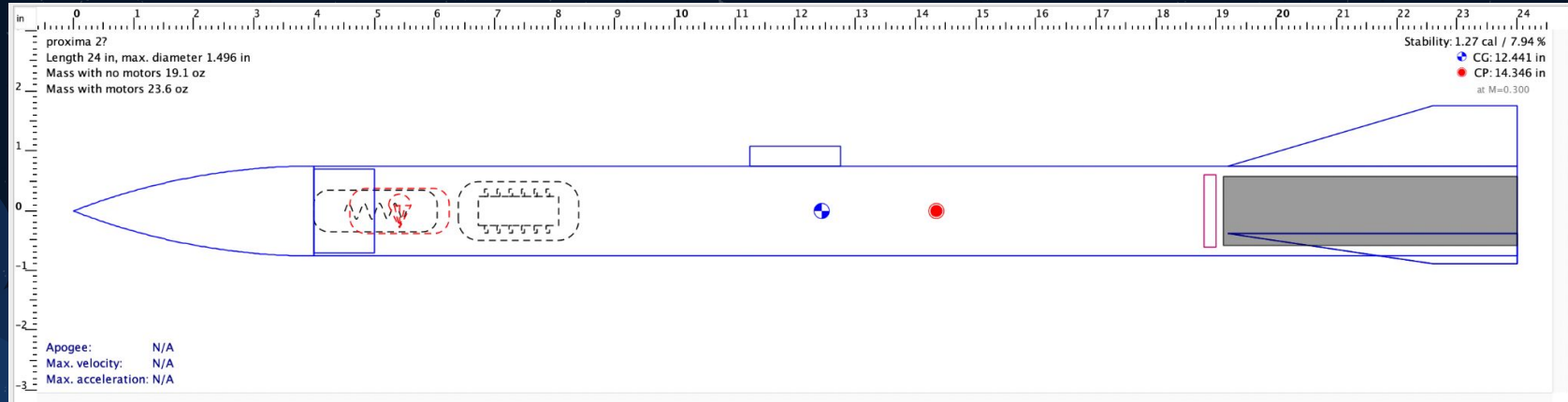




OVERVIEW OF PHYSICAL PROPERTIES

- Length: 24 in
- Total Mass: 1.475 lbs (0.669 kg)
- Stability: 1.27 caliber
- Apogee: 3552 ft
- Peak Velocity: 580 ft/s
- Peak Acceleration: 465 ft/s²
- CP = 14.346 in ; CG = 12.441 in

$$Stability = \frac{CP - CG}{outer\ diameter\ of\ body\ tube} = [cal]$$

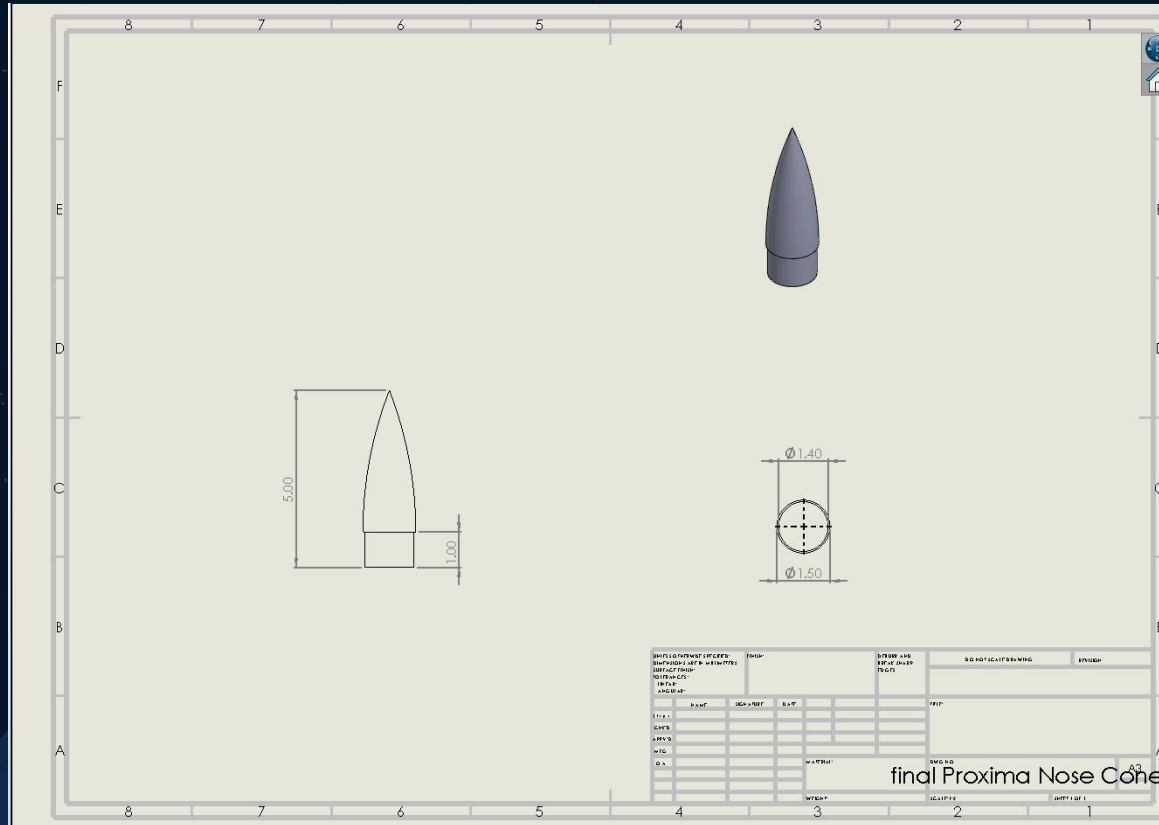


DESCRIPTION OF COMPONENTS



NOSE CONE

1. Nose Cone (0.62 oz)
 - a. L: 4 in + 1 in shoulder
 - b. PLA
 - c. Ogive Shape
 - d. Friction Fit to body

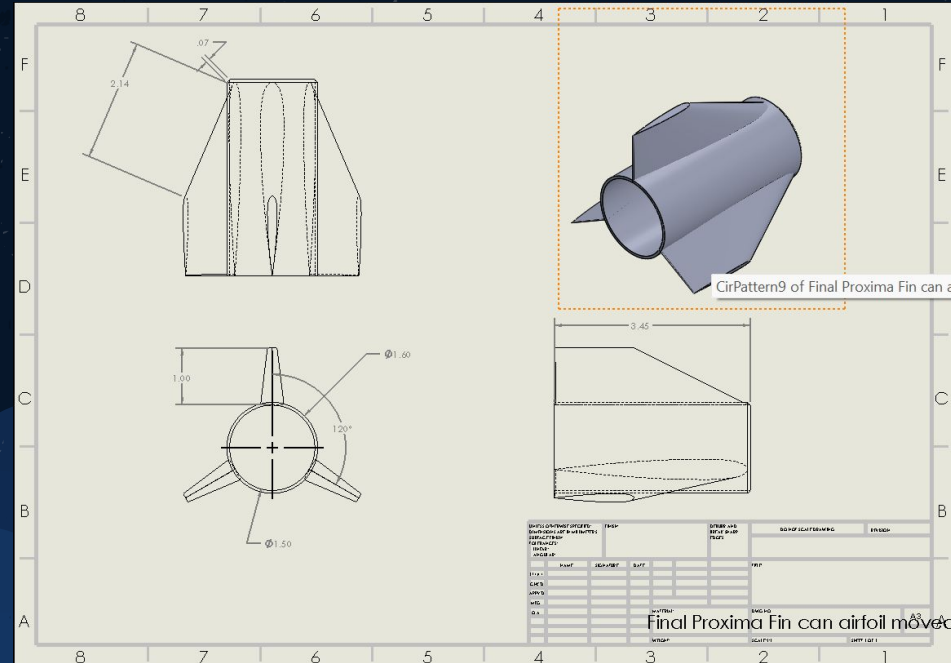




FINCAN

3 Fins

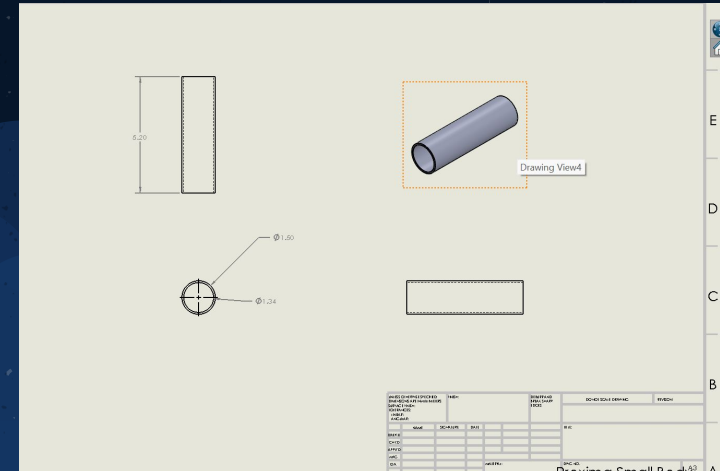
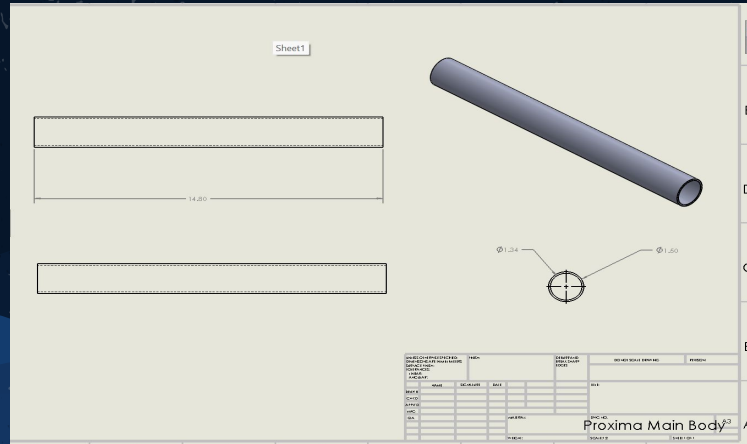
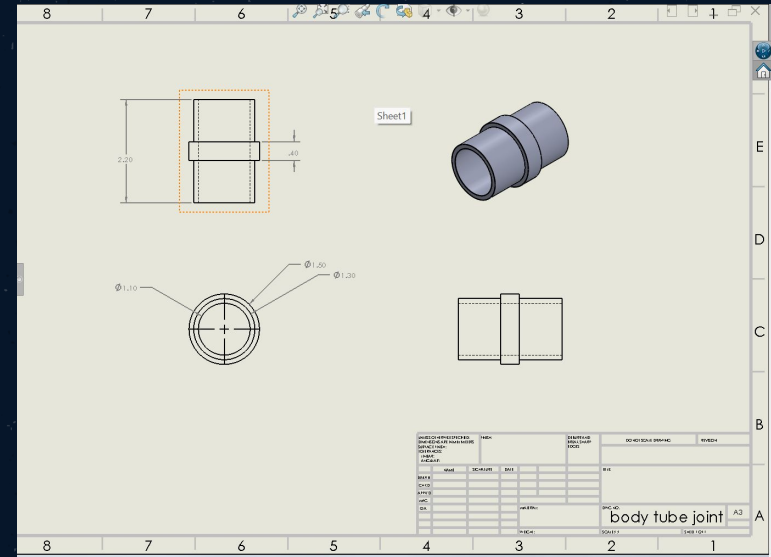
- Trapezoidal fin set
- NACA 63-212 airfoil
- L x H : 1.6 in x 0.87 in
- Thickness: 0.08 in
- 3D printing fin that will be friction fit onto body + poxy





BODY TUBE

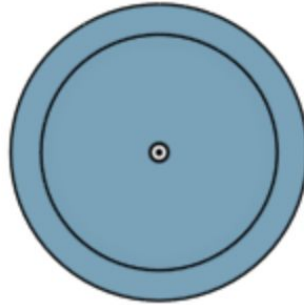
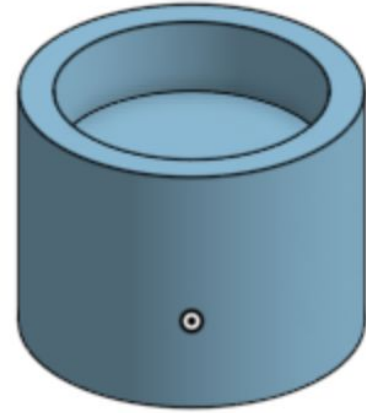
- Body Tube (4.1 oz)
 - D x L: 1.496 in x 20 in
 - 38mm Blue Tube
 - Split Body Tube
 - Parachute deploy from the center of the body
 - 3D printed Joint
 - Poxo on the Bottom
 - Friction Fit Top





PARACHUTE

- Ripstop Nylon from Apogee Rockets
- Canopy diameter of (24")
- Drag Coefficient 0.800
- Attached inside the body tube through the bolts
- 3D printing cylinder + attaching eye-bolt





MOTORS

- Blue Thunder [G80T - 13] Motor (AeroTech)
 - Total impulse of the G-class motor: 136 Ns
 - Diameter: 1.14 in
 - Length: 4.88 in
 - Avg. Thrust: 78N
 - Max Thrust: 102N
 - Burn Time: 1.71s
 - Launch Mass: 4.52 oz
 - Empty Mass: 2.32 oz



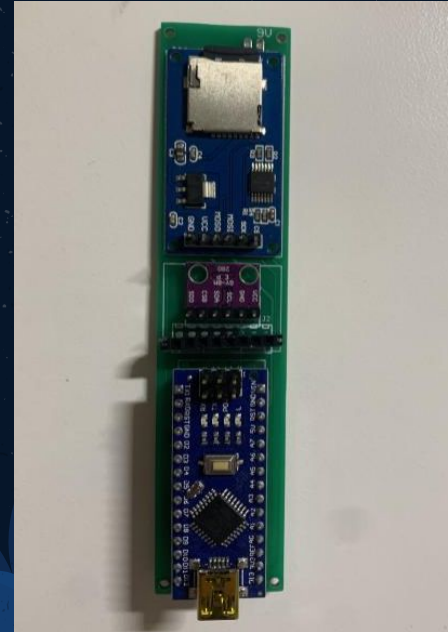
★ OLD AVIONIC COMPONENTS

- PCB
- MINI ARDUINO NANO
- pressure/altitude/temperature sensor (BMP 388)
- IMU/gyroscope/accelerometer/magnetometer (BN0055)
- Battery



★ NEW AVIONIC COMPONENTS

- Protoboard
- MINI ARDUINO NANO
- pressure/altitude/temperature sensor (BMP 388)
- IMU/gyroscope/accelerometer/magnetometer (BN0055)
- Battery



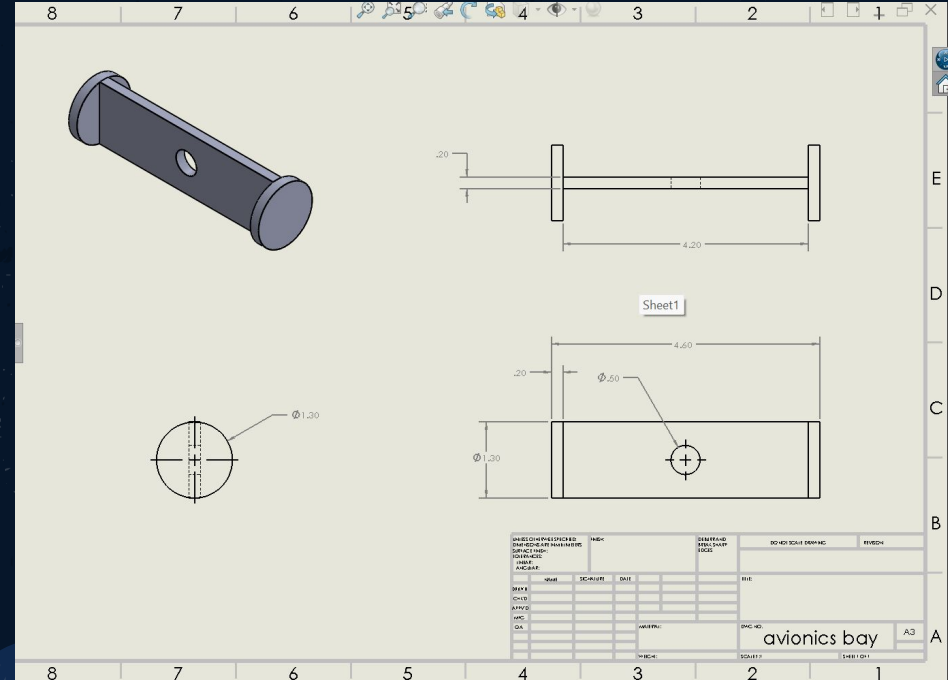
★ AVIONICS: WEIGHT AND COST

Arduino Nano	\$9.99	<u>Amazon Link</u>
Micro SD Card Module	\$6.99	<u>Amazon Link</u>
Micro SD Card	\$8.49	<u>Amazon Link</u>
BMP 280	\$5.99	<u>Amazon Link</u>
9V Battery Clip	\$7.99	<u>Amazon Link</u>
Jumper Wires	\$6.98	<u>Amazon Link</u>
GY-86 Flight Control Sensor	\$23.39	<u>Arduino Tech</u>
Total	\$69.82	



AVIONICS BAY

- Diameter - 1.30 in.
- Length- 4.60 in.
- Hole diameter- 0.5 in.
- 3D printed out of PETG
- Friction fit on top half of rocket

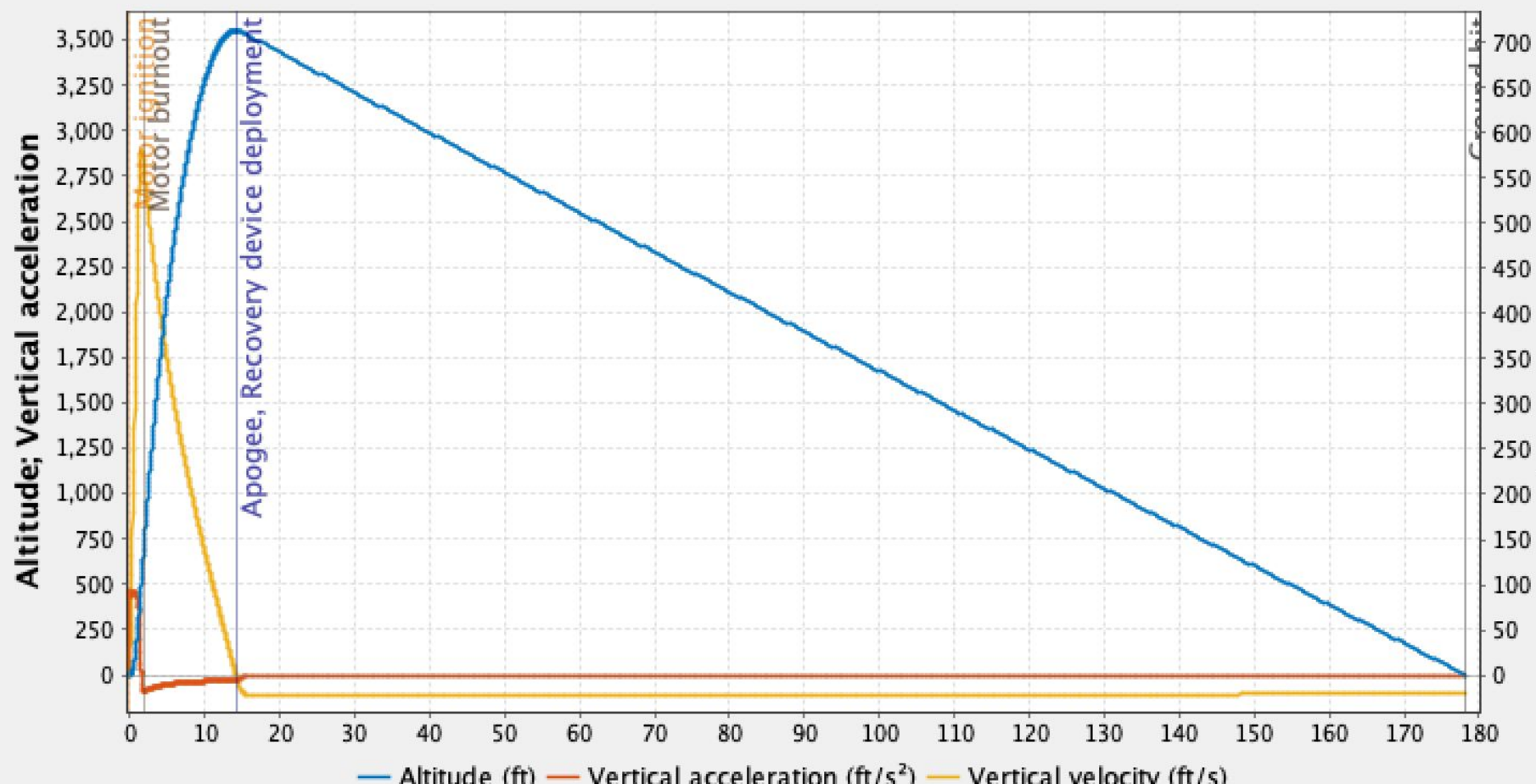


SIMULATION DATA



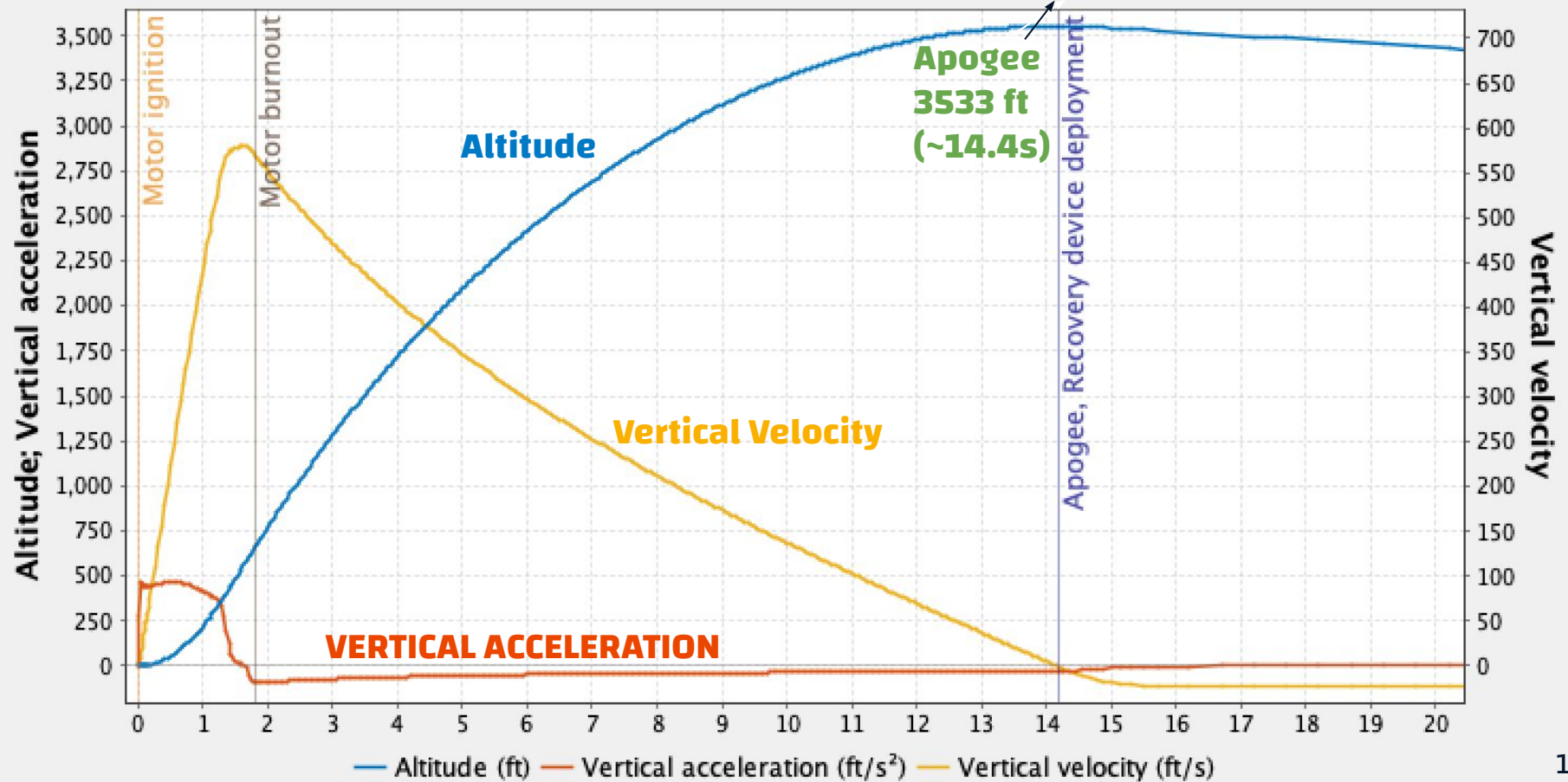
Simulation 4

Vertical motion vs. time

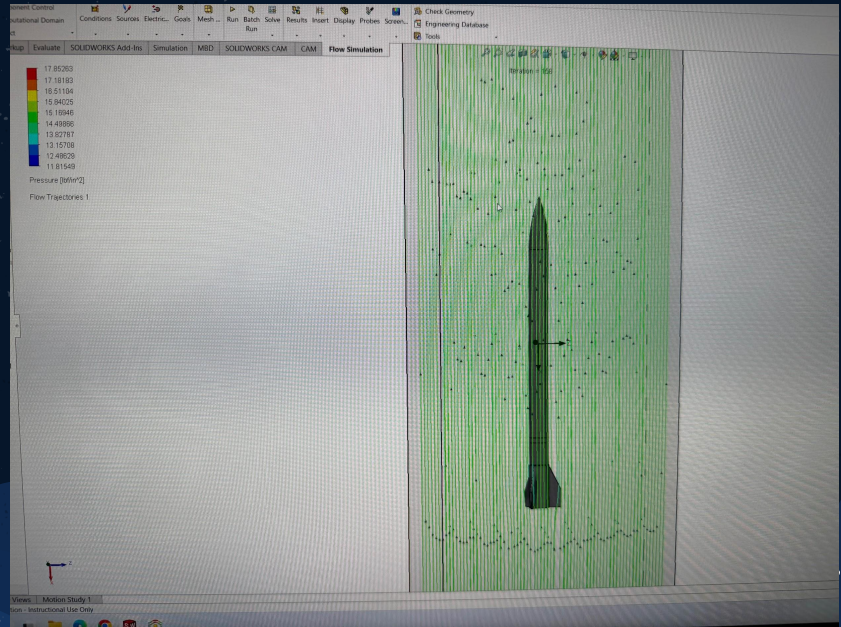
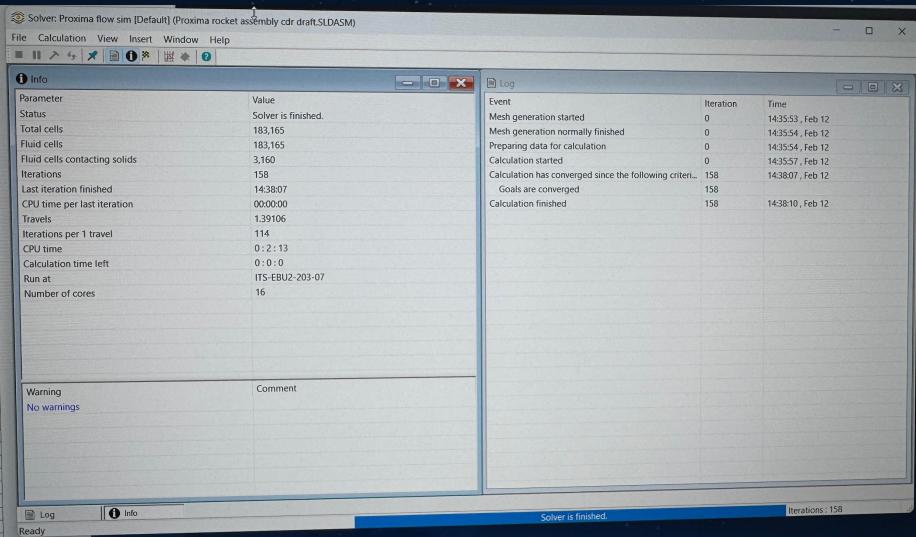


Simulation 4

Vertical motion vs. time



FLOW SIMULATION



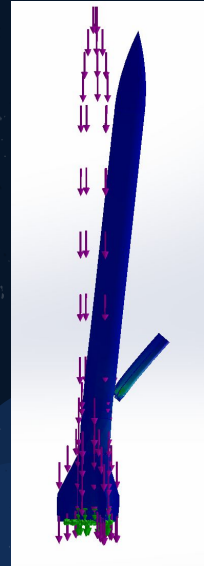
PRESSURE SIMULATION: STRAIN

0.1082 lbf

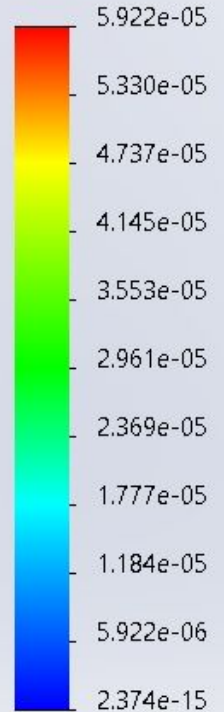
MAX = $5.922\text{e-}05$

MIN = $2.374\text{e-}15$

Model name: Proxima rocket assembly cdr draft
Study name: Rocket Static(-Default-)
Plot type: Static strain Strain1
Deformation scale: 4,953.4



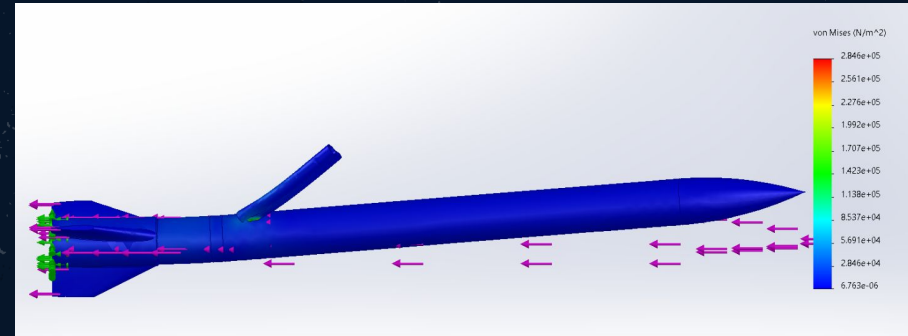
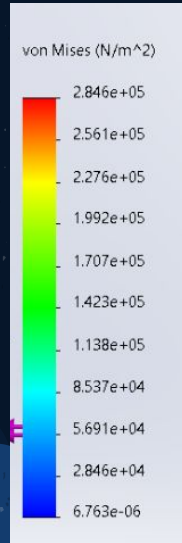
ESTRN



PRESSURE SIMULATION: STRESS

MAX = 2.846×10^5 (N/m²)

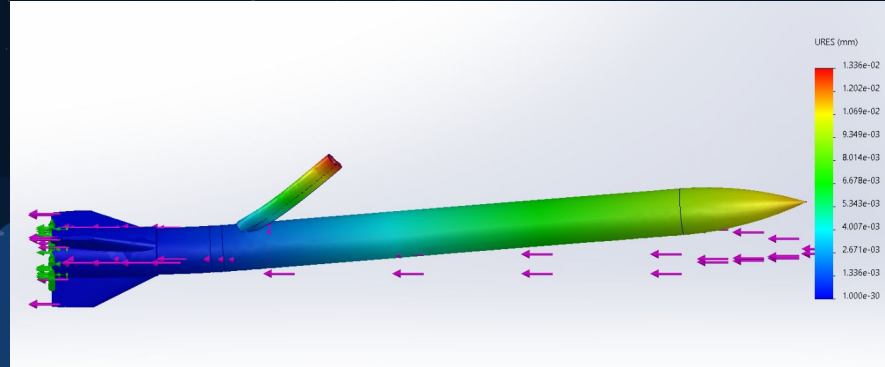
MIN = 6.763×10^{-6} (N/m²)



PRESSURE SIMULATION: DISPLACEMENT

MAX = $1.336e-02$ (mm)

MIN = $1.000e-30$ (mm)



TOTAL COST ESTIMATION



★ Total Cost Estimation

- Fins: Printed in-house
- Nose Cone: Printed in-house
- Body Tube: \$ 21.76
- Motor: ~ \$40
- Parachute: \$10.28
- Avionics: \$55

Item Name	Item Cost (\$)
Fins	\$0
Nose Cone	\$0
Body Tube	\$21.76
Motors	\$40
Parachute	\$10.28
Avionics	\$55
Total:	\$127



RPL

ROCKET PROPULSION LABORATORY

**AD ASTRA
PER ASPERA**